

8	(a) two waves (of the same kind) travelling in opposite directions overlap waves have same frequency/wavelength and speed	B1 B1	[2]
	(b) (i) $T = 0.8(\text{ms})$	C1	
	$f = 1 / (0.8 \times 10^{-3}) = 1250(\text{Hz})$	A1	[2]
	(ii) microphone is moved from plate to loudspeaker or vice versa wavelength is the twice the distance between adjacent maxima or minima (seen on c.r.o.)	B1 B1	[2]
	(iii) $v = f\lambda$	C1	
	$= 1250 \times 0.26$		
	$= 330 (325) \text{ms}^{-1}$	A1	[2]